

designing customised blockchains, each tailored to specific use cases. Use cases for Polkadot span a wide range of industries, including finance, supply chain, healthcare, and more. By addressing the challenges of scalability, interoperability, and customisation, Polkadot is playing a pivotal role in advancing the capabilities of decentralised applications and the broader blockchain ecosystem.

Cardano

URL: cardano.org

Cardano is a blockchain platform that aims to provide a more secure and scalable infrastructure for the development of smart contracts and decentralised applications. It uses the Haskell programming language for smart contract development. Launched with a research-driven methodology, Cardano distinguishes itself by incorporating academic peer-reviewed research into its development process. The platform operates on a layered architecture, separating the settlement and computation layers, aiming to enhance scalability and flexibility. Cardano's native cryptocurrency, ADA, facilitates secure and transparent transactions within its ecosystem. With a focus on sustainability, interoperability, and regulatory compliance, Cardano seeks to address challenges prevalent in the blockchain space. The use cases for Cardano span a variety of sectors, including finance, identity verification, supply chain management, and governmental services. Its commitment to academic rigour and a comprehensive development strategy positions Cardano as a promising player in the blockchain realm.

NEO

URL: neo.org

NEO is a blockchain platform that supports smart contracts. Developers can use languages like C#, Java, and Python for creating smart contracts on the NEO platform. Founded with

the vision of establishing a more scalable and user-friendly blockchain ecosystem, NEO supports multiple programming languages, including C#, Java, and Python, making it accessible to a broad range of developers. One of NEO's key strengths lies in its ability to facilitate the creation of decentralised applications (dApps) and smart contracts in a highly efficient and secure manner. NEO's use cases extend across various industries, from finance to supply chain management, where its advanced features, such as a Byzantine fault-tolerant consensus mechanism and a unique delegated Byzantine Fault Tolerance (dBFT) consensus algorithm, contribute to the platform's reliability and robustness.

Avalanche

URL: avalabs.org

Avalanche supports the creation of interoperable and highly scalable applications. Recognised for its unique consensus protocol called Avalanche Consensus, the platform distinguishes itself by providing high-throughput, near-instant transaction finality, and robust security. Avalanche stands out as a leader in the blockchain space due to its ability to support custom blockchain networks and facilitate the creation of decentralised applications (dApps). The platform's flexibility, scalability, and interoperability contribute to its growing popularity among developers and enterprises.

Tezos

URL: tezos.com

Tezos is a self-amending blockchain platform that supports smart contracts and decentralised applications. It utilises the Michelson programming language for smart contract development.

Algorand

URL: algorand.com

Algorand is a blockchain platform that focuses on high-performance and decentralised applications. It uses a unique consensus algorithm called Algorand Consensus.

Tron

URL: tron.network

Tron is a blockchain platform known for its emphasis on content and entertainment applications. It supports the creation of decentralised applications and smart contracts.

EOS

URL: eos.io

EOS is a blockchain platform that aims to provide a user-friendly environment for building decentralised applications. It uses a delegated proof-of-stake consensus mechanism.

Exploring these platforms allows developers to choose the one that best aligns with their project requirements, programming preferences, and scalability needs. The blockchain landscape continues to evolve, and each platform contributes to the diversity and innovation within the decentralised application development space.

The scope for research and development in smart contract programming is vast. New platforms are emerging, each with its unique features and advantages. Researchers can explore areas such as scalability, privacy, and sustainability within smart contract platforms. Interoperability between different blockchain networks is another exciting avenue for development. By contributing to the open source community, developers can shape the future of decentralised applications and smart contract programming. **END** 

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